

AMENDMENT NO. 1 APRIL 1985

TO

IS : 6848-1979 SPECIFICATION FOR LEAD-ACID BATTERIES FOR TRAIN LIGHTING AND AIRCONDITIONING SERVICES

(First Revision)

(Page 5, Table 1, Sl No. 7) — Delete together with all entries against it.

(Page 8, Fig. 2, Note 1) — Substitute ' 0.01 mm ' for ' 0.025 mm '.

(Page 12, clause 5.1.2) — Substitute the following for the existing clause:

'5.1.2 Acceptance Test — The following tests shall be conducted as acceptance tests:

- a) Capacity test at 10 h rate (5.4), and
- b) Checking of dimensions, mass, markings and workmanship (5.14).

(Page 12, clause 5.1.2) — Add the following new clause after 5.1.2 :

'5.1.3 Sampling Scheme and Criteria for Acceptance

5.1.3.1 A recommended sampling scheme and criteria for the acceptance of the lot for various lot sizes is given in Appendix G. '

(Page 12, clause 5.2) — Add the following at the end of the clause:

'Use of a digital ammeter/voltmeter is preferable for testing. Chart recorders shall be used for life cycle testing.'

(Page 16, clause 5.9.8) — Substitute the following for the existing matter:

'5.9.8 The number of life units shall not be less than 6. '

(Page 21, Appendix D) — Add the following Note after sequence of type tests:

'NOTE — The cell shall be covered by type approval certificates from an appropriate authority. Significant variations in the design shall be covered by separate type approval certificates. The cell of new design shall be proto-type tested for which samples offered by the manufacturer shall be accepted. For type test samples drawn from mass production at random shall be retested after 5 years. In case of unsatisfactory performance of cells, type testing can be repeated earlier at the discretion of the approving authority.'

(Page 22, Appendix F) — Add the following as Appendix G after Appendix F:

‘ APPENDIX G

(Clause 5.1.3.1)

SAMPLING SCHEME AND CRITERIA FOR CONFORMITY FOR ACCEPTANCE TESTS

G-1. LOT

G-1.1 All the batteries of the same type, design and rating manufactured from the same material by the same factory under similar conditions of production shall constitute a lot.

G-1.2 The batteries to be selected from each lot shall depend upon the size of the lot and shall be in accordance with col (1) and (2) of Table 2:

TABLE 2 SAMPLE SIZE, ACCEPTANCE AND REJECTION NUMBERS
(Clauses G-1.2 and G-2.1)

LOT SIZE	FIRST STAGE	SECOND STAGE		ACCEP- TANCE NUMBER	FIRST REJEC- TION NUMBER	SECOND REJEC- TION NUMBER
(N)	(n)	(n)	(2n)	(a)	(r ₁)	(r ₂)
(1)	(2)	(3)	(4)	(5)	(6)	(7)
Up to 50	2	—	2	0	1	—
51 „ 300	3	—	3	0	1	—
301 „ 500	5	5	10	0	2	2
501 „ 1 000	8	8	16	0	2	2
1 001 and above	13	13	26	0	3	4

NOTE 1 — The sampling plan given in the table envisages that the lots containing about 4 percent defective batteries will be accepted most of the time.

NOTE 2 — For lots of size up to 300, the decision regarding acceptance or rejection of the lots shall be taken at the first stage only.

G-1.2.1 These batteries shall be selected from the lot at random. In order to ensure the randomness of selection, procedures given in IS : 4905-1968* may be followed.

G-2. NUMBER OF TESTS AND CRITERIA FOR CONFORMITY

G-2.1 All the batteries selected at the first stage in accordance with col (1) and (2) of Table 2 shall be subjected to the acceptance tests.

*Methods for random sampling.

A battery failing to satisfy the requirements of any one of the acceptance tests shall be termed as defective. The lot shall be considered as conforming to the requirements of acceptance tests if no defective is found and shall be considered as non-conforming to the requirements if the number of defectives is equal to or greater than the first rejection number (r_1). If the number of defectives is greater than the acceptance number (a) but less than the first rejection number (r_1), the second sample of the same size as taken in the first stage shall be selected from the lot at random and subjected to the acceptance tests. The number of defectives in the first sample and the second sample shall be combined. If the combined number of defectives is less than the second rejection number (r_2), the lot shall be considered as conforming to the requirements of acceptance tests; otherwise not.

(ETDC 11)